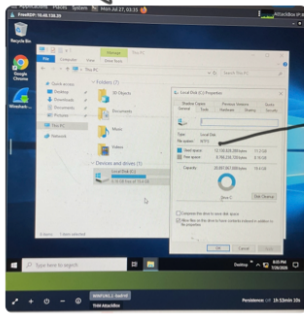


Task 4:- The File System

The file system used in modern version of windows is New technology file system or **NTFS**.



→ NTFS

Before NTFS, there was FAT16/FAT32 (File Allocation Table) and HPFS (High performance file system). You still see FAT partitions in use today. You typically see FAT partitions in USB devices, MicroSD cards, etc.

* NTFS known as **Journaling file system**. In case of failure, the file system can automatically repair folders/files on disk using information stored in log file.

NTFS Addresses many of limitations of previous file systems, such as:

- * Support files larger than 4GB
- * Set specific permissions on folders and files
- * Folder and file compression
- * Encryption (Encryption file system or EFS)

If you're running windows, what is the file system your windows installation is using?

You can check properties right click of drive your operating system is installed on.

Let's speak briefly on some features that are specific to NTFS.

On NTFS volumes, you can set permissions that grant or deny access to files and folders.

The permissions are:

- * Full control
- * Modify
- * Read and Execute
- * List folder contents
- * Read
- * Write

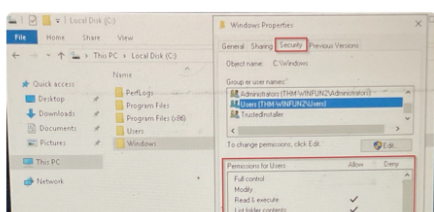
Permission	Meaning for Folders	Meaning for Files
Read	Permits viewing and listing of files and subfolders	Permits viewing or accessing of the file's contents
Write	Permits adding of files and subfolders	Permits writing to a file
Read & Execute	Permits viewing and listing of files and subfolders as well as executing of files inherited by files and folders	Permits viewing and accessing of the file's contents as well as executing of the file
List Folder Contents	Permits viewing and listing of files and subfolders as well as executing of files inherited by folders only	N/A
Modify	Permits reading and writing of files and subfolders; allows deletion of the folder	Permits reading and writing of the file; allows deletion of the file
Full Control	Permits reading, writing, changing, and deleting of files and subfolders	Permits reading, writing, changing and deleting of the file

How can we set permissions for file or folder?

* Right click file or folder you want to check for permissions

* From context menu, select **properties**.

* Within properties, click on **Security** tab.



* Refer the Microsoft documentation to get better

understanding of NTFS permissions for **Special permissions**.



* Another feature of NTFS is Alternate Data Streams (ADS)

* Alternate Data Streams (ADS) is file attribute specific to windows NTFS (New Technology file System)

* Every file has at least one data stream (\$DATA), and ADS allows files to contain more than one stream of data.

From security perspective, malware writers have used ADS to hide data.

Ex:- If you download a file from internet, there are identifiers written to ADS to identify that file was downloaded from the internet.

Quiz:-

1) What is the meaning of NTFS?

A) New Technology File System.

Task 5:- The windows\System32 folders

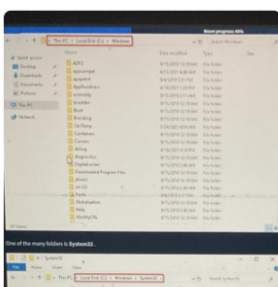
The windows folder (C:\windows) is traditionally known as folder which contain windows operating system. The folder doesn't have to reside in c drive necessarily, it can reside in any other drive and technically can reside in different folder.

There is where environment variables, more specifically system environment variables.

Even though not discussed yet, the system environment variable for window directory is

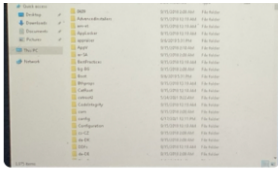
%windir%.

per Microsoft, Environment variables store information about operating system environment. This information includes details such as operating system path, The number of processors used by operating system, and location of temporary folders.



* The System32 folder holds the important files that are critical for OS.

You should proceed with extreme caution when interacting with this folders. Accidentally deleting any files & folders within C:\Windows\



Can render the windows OS Inoperational.

Quiz:-

What is system variable for windows folder?

A) %windir% → when you type on Search it will go navigate to System 32.

Task 6:- User Accounts, Profiles and permissions

User Accounts can be one of two types on typical local windows System: Administrator and Standard user.

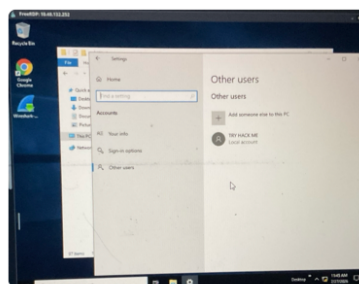
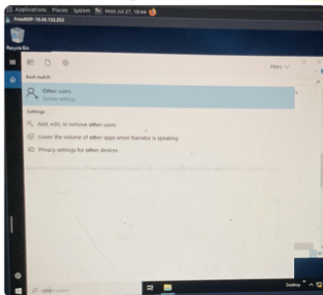
The user account type will determine what actions user can perform on that specific windows System.

→ An Administrator can make changes to System: - add users, delete users, modify groups, modify Settings on System.

→ A Standard users can only make changes to folders/files attributed to the user & can't perform System level changes, such as install programs.

You are logged in as Administrator. There are several ways to determine which user account exist on System.

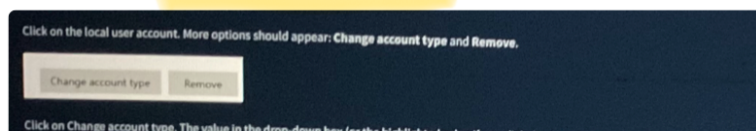
One way to Start menu and type other users. A shortcut to System Settings > other users should appear.

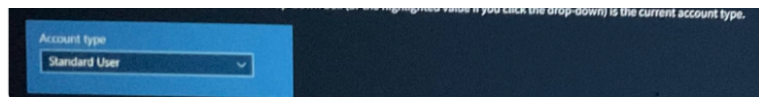


Since, you're the administrator, you see an option Add Someone else to this PC.

Note 1:- A Standard user will not see this option.

Click on local user account. More options should appear: change account type and remove.

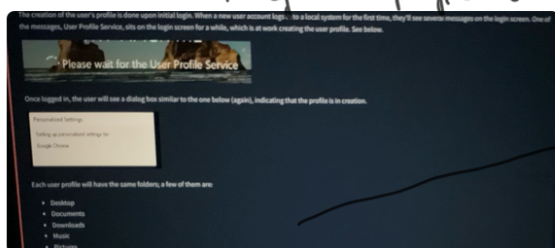




When user Account is created, a profile is created for the user. The location for each user profile folder will fall in **C:\Users**

Ex:- The user profile folder for user Account Man will be **C:\Users\Man**

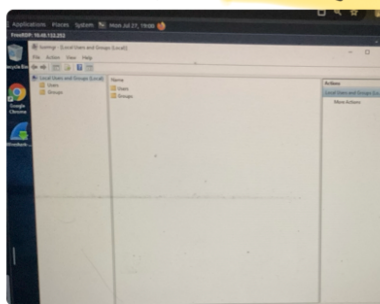
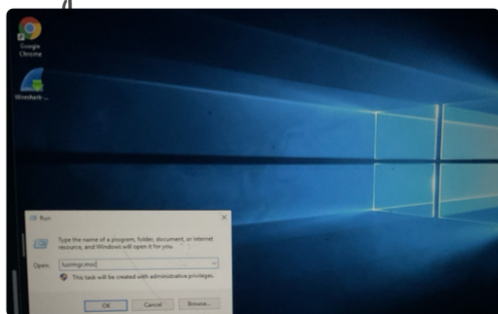
The Creation of user profile is done upon initial login.



→ Desktop, Documents, Downloads, Music, Pictures.

Another way to access this information, and then some is using **local user and Group management**.

Right click on Start menu and click Run 'Type **lusrmgr.msc**



→ you could see two folders: users and groups.

If you click on Groups, you see all names of local groups along with brief description for each group.

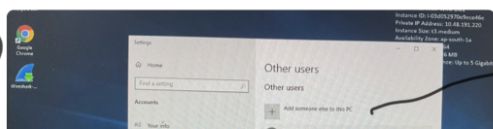
Each group has permissions set to it, and users are assigned/added to groups by Administrator. When user is assigned to group, the user inherits the permissions of that group. A user can be assigned to multiple groups.

Note:- If you click on Add Someone else to this PC from other users, it will open **local users and management**.

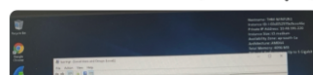
Quiz:-

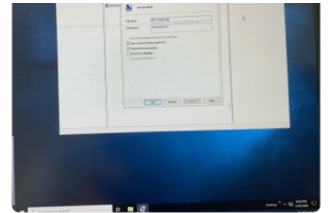
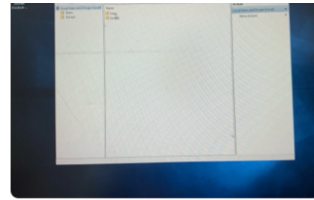
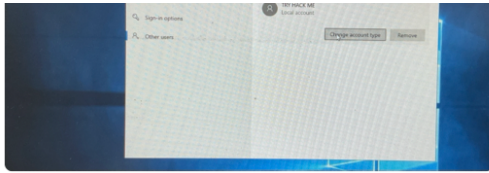
1) What is the name of other user account?

A)



→ Add Someone else to this PC and then select users

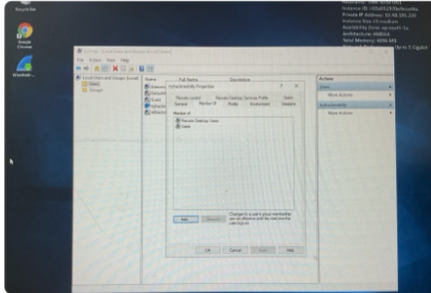




to yhack me billy

2) What groups is this user member of?

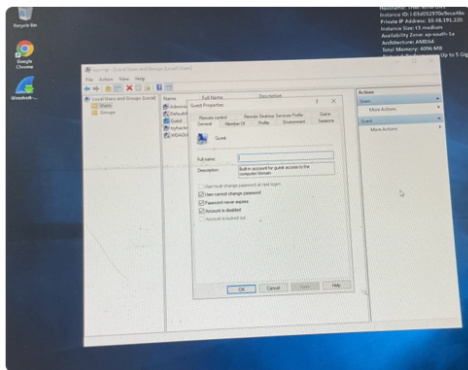
A)



Remote Desktop users, Users

3) What Built in Account is for guest access to this computer?

A) Guest



4) What is the account description?

A) window\$Fun!

